Kantipur City College

Department of Computer Engineering

Semester IV/ Probability and Statistics

Lab Sheet No.10

Title: Confidence Interval estimation.

Objectives: To find confidence interval estimate for population mean and proportion.

Theory:

Confidence interval for population mean:

$[\overline{X} \pm Z_{\propto}$ S.E. $(\overline{X})$]

Where,

S.E $(\overline{X}) = \frac{\sigma}{\sqrt{n}}$ if population is infinite.

S.E $(\overline{X}) = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$ if population is finite.

Confidence interval for population proportion:

$[p \pm Z_{\propto}$ S.E. (p)]

Where,

S.E (p) = $\sqrt{\frac{pq}{n}}$ if population is infinite.

S.E (p) = $\sqrt{\frac{pq}{n}} \sqrt{\frac{N-n}{N-1}}$ if population is finite.

Lab work:

1. A random sample of 50 gave a mean of 7.5 kg and standard deviation of 1.5 kg. Find 95% and 99% confidence limit for the population mean.

Using Excel:

	A	B	C
1	**Data**		**Formula**
2	n=	50	
3	sample mean=	7.5	
4	sample standard deviation=	1.5	
5	confidence coefficient (1-α)=	0.95	
6	α=	0.05	=1-B5
7	α/2=	0.025	=B6/2
8	Zα/2=	1.959963985	=ABS(NORMSINV(B7))
9	Standard err0r of mean=	0.212132034	=B4/SQRT(B2)
10			
11	**Calculation of 95% C.I for mean:**		
12	Lower confidence limit=	7.084228853	=B3-B8*B9
13	Upper confidence limit=	7.915771147	=B3+B8*B9
14			
15	Hence required 95% confidence limit for mean is	(7.0842289,7.9157711)	

2. From a population of 540, a sample of 60 individuals is taken. From this sample, the mean is found to be 6.2 and standard deviation 1.368. Construct 98% confidence interval for the mean.

Using Excel:

	A	B	C
1	**Data**		**Formula**
2	n=	60	
3	N=	540	
4	sample mean=	6.2	
5	sample standard deviation=	1.368	
6	confidence coefficient (1-α)=	0.95	
7	α=	0.05	=1-B5
8	α/2=	0.025	=B6/2
9	Zα/2=	1.959964	=ABS(NORMSINV(B7))
10	Standard err0r of mean=	0.166662	=B5/SQRT(B2)*SQRT((B3-B2)/(B3-1))
11			
12	**Calculation of 95% C.I for mean:**		
13	Lower confidence limit=	5.873348	=B4-B9*B10
14	Upper confidence limit=	6.526652	=B4+B9*B10

3. A random sample of 500 oranges was taken from a large consignment and it was observed that 65 were found to be bad. Find 99% confidence interval estimate for the proportion of bad oranges.

Using Excel:

	A	B	C
1	**Data**		**Formula**
2	n=	500	
3	No. of bad oranges(X)=	65	
4	p=	0.13	=B3/B2
5	q=	0.87	=1-B4
6	Standard error of p=	0.01504	=SQRT(B4*B5/B2)
7	confidence coefficient(1-α)=	0.99	
8	α=	0.01	=1-B7
9	α/2=	0.005	=B8/2
10	zα/2=	2.575829	=ABS(NORMSINV(B9))
11			
12	**99% C.I. for population proportion**		
13	Lower confidence limit=	0.09126	=B4-B10*B6
14	Upper confidencelimit=	0.16874	=B4+B10*B6

4. A factory is producing 50000 pairs of shoes daily, 500 pairs of shoes are drawn as sample in which 2% were found to be defective. Assign 95% limits within which proportion of defective pair lies.

	A	B	C
16			
17			
18	Data		Formula
19	N=	50000	
20	n=	500	
21	p=	0.02	
22	q=	0.98	=1-B21
23	Confidennce coefficient(1-α)=	0.95	
24	zα/2=	1.959964	
25	standard error of p=	0.00623	=SQRT((B21*B22)/B20)*SQRT((B19-B20)/(B19-1))
26			
27	**95% confidence limit for proportions**		
28			
29	Lower cconfidence limit=	0.00779	=B21-B24*B25
30	upper confidence limit=	0.03221	=B21+B24*B25

Assignments;

1. If sample mean is 20, population standard deviation is 3 and the sample size is 64, find the interval estimate of the population mean. (α=5%)

2. In a sample survey of 1000 housewives in a city 23% prefer a particular brand of pressure cooker. Find 99% confidence limits for the percentage of all housewives in the city preferring that brand of pressure cooker.